

Meng-Chieh (Jeremy) Lee

CONTACT  
INFORMATION

Website: <https://mengchillee.github.io/>  
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EDUCATION

**Carnegie Mellon University**  
Ph.D., Computer Science Department  
**Advisor:** Christos Faloutsos

Pittsburgh, PA  
Aug 2021 - May 2025

WORK  
EXPERIENCE

**Amazon Web Services**  
Applied Scientist

Santa Clara, CA  
June 2025 - Present

- **Automated Data Improvement for Model Fine-Tuning:** Initiated and spear-headed cross-functional deployment of a data improvement pipeline driven by automated error analysis. Partnered with engineering teams to productionize the system, driving a 22% performance improvement in the fine-tuned model with improved data.
- **Self-Evolving Prompt Optimization Agent:** Devised an MCTS-based agent to resolve manual iteration bottlenecks. Pioneered structural dimensionality reduction by modeling meta-prompts as permutation-invariant instruction sets to bound the search space, outperforming human baselines by 25% and slashing token costs by 61%.
- **Multi-Agent Optimization Orchestration:** Built a generalized framework that orchestrates parallel agents to optimize objectives, eliminating the need for task-specific pipelines. Applied meta-memory and test-time scaling to optimize an LLM for translation, closing 50% of the performance gap to the best-performing LLM baseline.
- **Scalable Shadow Testing Framework:** Architected a lightweight template matching pipeline to enable cost-effective shadow testing for an LLM service. Replaced expensive evaluation methods, successfully safeguarding production deployments while boosting relative success rates by 90% and reducing compute costs by 3x.
- **Multi-Objective Prompt Optimization:** Identified fundamental limitations in the team’s baseline approach. Architected and deployed a structural overhaul to resolve prompt content interference, successfully co-optimizing complex objectives to unlock simultaneous relative gains of 11.5% and 12.3% for a flagship product.
- **Research Mentorship:** Conceived and pioneered an efficient LLM reasoning research, personally building an oracle PoC that matched extended models with 80% fewer tokens. Currently directing an intern to scale this via fine-tuning. Previously co-mentored an intern on Text-to-SQL RL agents, yielding an ACL 2026 publication.

**Amazon**  
Applied Scientist Intern

Bay Area, CA  
May - Aug

- **2024:** Developed *HybGRAG*, an agentic framework combining retrieval in text databases and knowledge graphs, improving top-1 retrieval accuracy by 51% while reducing API calls by 97%; accepted to ACL 2025 Main.
- **2023:** Developed *NetInfoF*, a training-free framework for measuring and exploiting usable graph information, outperforming GNN baselines on 11 of 12 link-prediction tasks; selected as an ICLR 2024 Spotlight.
- **2022:** Identified a systemic server-side time-series anomaly affecting all core services and developed a mitigation strategy to address recurring system failures.

## SELECTED PUBLICATIONS

1. Hua, H., Han, Z., Shen, Z., **Lee, M.C.**, Guan, S., Zhu, Q., Jeoung, S., Chen, Y., Bai, Y., Wang, S., Ioannidis, V.N., Rangwala, H., “SQL-Trail: Multi-Turn Reinforcement Learning with Interleaved Feedback for Text-to-SQL”. *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
2. **Lee, M.C.**, Zhu, Q., Mavromatis, C., Han, Z., Adeshina, S., Ioannidis, V.N., Rangwala, H., Faloutsos, C., “HybGRAG: Hybrid Retrieval-Augmented Generation on Textual and Relational Knowledge Bases”. *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. (Main).
3. Deforce, B., **Lee, M.C.**, Baesens, B., Serral Asensio, E., Yoo, J., and Akoglu, L., “TSA on AutoPilot: Self-tuning Self-supervised Time Series Anomaly Detection”. *SIAM International Conference on Data Mining (SDM)*, 2025.
4. **Lee, M.C.**, Shekhar, S., Yoo, J., and Faloutsos, C., “Discovery and Exploitation of Generalized Network Effects”. *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2024.
5. **Lee, M.C.\***, Zhao, L.\*, and Akoglu, L., “Descriptive Kernel Convolution Network with Improved Random Walk Kernel”. *ACM Web Conference (WWW)*, 2024.
6. **Lee, M.C.**, Yu, H., Zhang, J., Ioannidis, V.N., Song, X., Adeshina, S., Zheng, D., and Faloutsos, C., “NetInfoF Framework: Measuring and Exploiting Network Usable Information”. *International Conference on Learning Representations (ICLR)*, 2024. (Spotlight).
7. Yoo, J.\*, **Lee, M.C.\***, Shekhar, S., and Faloutsos, C., “Less is More: SLIMG for Accurate, Robust, and Interpretable Graph Mining”. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023.
8. Vajiac, C., **Lee, M.C.**, Kulshrestha, A., Levy, S., Park, N., Olligschlaeger, A., Jones, C., Rabbany, R., and Faloutsos, C., “DELTASHIELD: Information Theory for Human-Trafficking Detection”. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 2023.
9. **Lee, M.C.\***, Shekhar, S.\*, Faloutsos, C., Hutson, T.N., and Lasemidis, L., “GEN<sup>2</sup>OUT: Detecting and Ranking Generalized Anomalies”. *IEEE International Conference on Big Data (BigData)*, 2021.
10. **Lee, M.C.\***, Vajiac, C.\*, Kulshrestha, A., Levy, S., Park, N., Jones, C., Rabbany, R., and Faloutsos, C., “INFOSHIELD: Generalizable Information-Theoretic Human-Trafficking Detection”. *37th IEEE International Conference on Data Engineering (ICDE)*, 2021.
11. **Lee, M.C.**, Zhao, Y., Wang, A., Liang, P.J., Akoglu, L., Tseng, V.S., and Faloutsos, C., “AUTOAUDIT: Mining Accounting and Time-Evolving Graphs”. *IEEE International Conference on Big Data (BigData)*, 2020.

(Full list of publications available on Google Scholar)

## IN THE NEWS

### Honeybee Health Monitoring

March 2025

- MSN
- CMU News

### INFOSHIELD for Human Trafficking Detection

May 2021

- WPXI
- CMU News